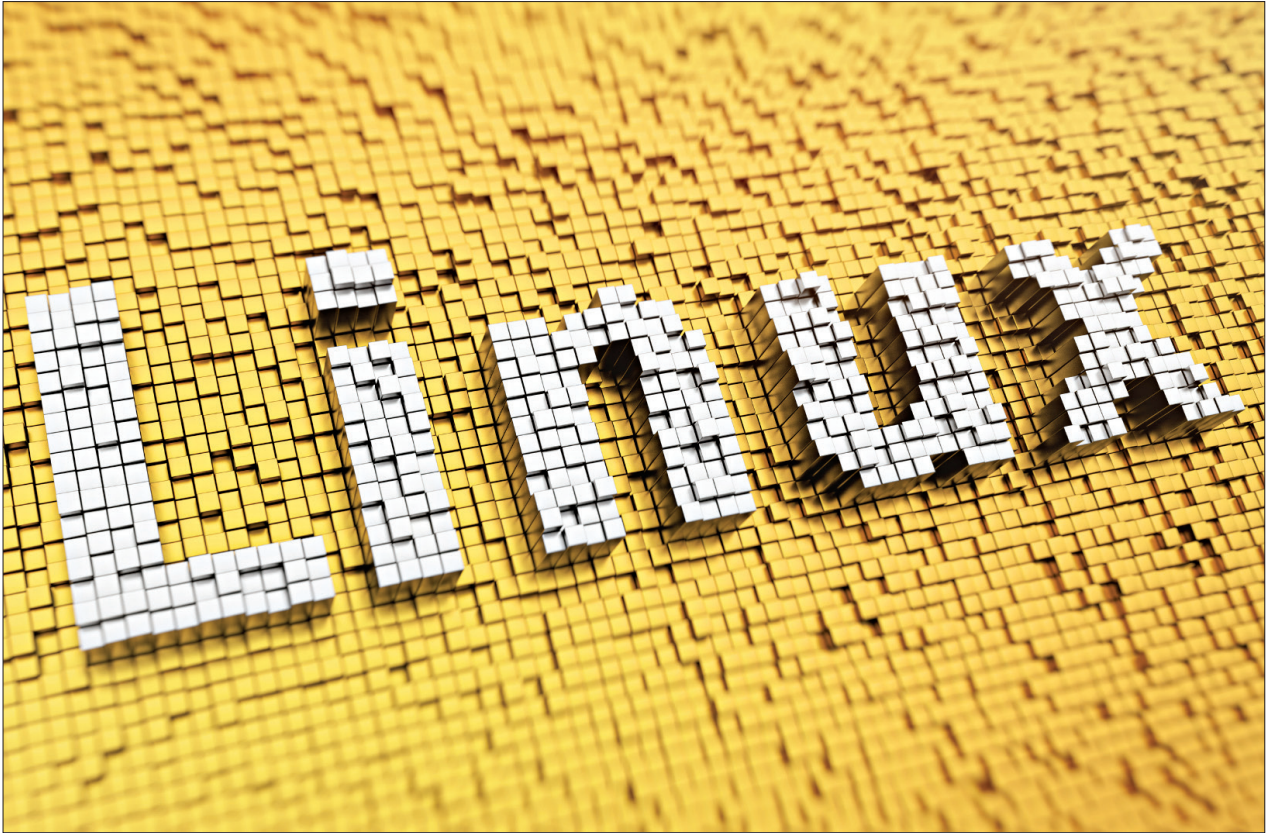




Introduction to Linux

The first steps

In a world where operating systems silently shape our digital experiences, Linux stands as a quiet revolution. Unlike mainstream systems that thrive behind proprietary walls, Linux is an open-source giant—free, adaptable, and deeply embedded in the infrastructure of modern computing. It powers everything from smartphones to supercomputers, yet remains elusive to the everyday user. But what exactly is Linux? How did it come to dominate the unseen layers of technology? To understand this, we must trace its roots to the bedrock of computing history.

What is Linux?

Origins: UNIX, GNU, and the Free Software Movement

Linux is more than an operating system; it is a philosophy born

from the UNIX lineage. The UNIX operating system, developed in the late 1960s at AT&T's Bell Labs, introduced key computing principles: multi-user environments, hierarchical file structures, and a modular design. UNIX's effectiveness made it highly sought after, but its licensing soon restricted access, prompting a counter-movement.

Enter Richard Stallman and the GNU Project. In the 1980s, Stallman envisioned a completely free and open UNIX-like system. His Free Software Foundation (FSF) spearheaded this cause, laying the groundwork for what would become the GNU operating system. However, one crucial element remained missing—a kernel.

That missing piece arrived in 1991 when Linus Torvalds, a Finnish computer science student, developed the Linux kernel. By merging the GNU software with Torvalds' kernel, a fully functional, free UNIX-like operating system emerged. Linux was